Functional Document

1. Introduction

The **TB and Lung Disease Diagnostic System** is an advanced medical imaging solution designed to address the global crisis of respiratory diseases. Pulmonary conditions such as Tuberculosis (TB), Pneumonia, and COVID-19 continue to be leading causes of morbidity worldwide, often exacerbated by a shortage of specialized radiologists in high-burden regions. This project introduces an automated diagnostic pipeline that utilizes state-of-the-art Deep Learning to analyze Chest X-ray (CXR) images. By providing a high-accuracy classification engine, the system serves as a critical decision-support tool, enabling healthcare providers to triage patients more efficiently and initiate life-saving treatments with greater speed and confidence.

2. Product Goal

The primary objective of this project is the development and deployment of a **Hybrid Deep Learning Model** that achieves clinical-grade diagnostic accuracy. Specifically, the goal is to:

- Successfully classify Chest X-rays into four distinct categories: **COVID-19, Normal, Pneumonia, and Tuberculosis**.
- Leverage a dual-backbone architecture—combining **EfficientNetB3** for scalable feature extraction and **DenseNet121** for dense connectivity—to ensure the model captures both global structural anomalies and local radiographic textures.
- Maintain a benchmark test accuracy of at least **95.5%**, with a specific focus on high recall for Tuberculosis to minimize the risk of false negatives in critical diagnostic scenarios.

3. Demography (Users, Location)

- **Target Users:**
    - **Radiologists and Pulmonologists:** Primary users who require a high-precision tool to validate their findings or to prioritize a massive influx of scans.
    - **General Practitioners (GPs) in Remote Clinics:** Users with limited specialized training in radiology who need an automated "second opinion" to assist in initial patient screening.

- **Laboratory Technicians:** Individuals responsible for the initial ingestion of patient data and X-ray imaging into the clinical system.

- **Location:**
  - **Primary Focus:** Public health institutions and private diagnostic centers in regions with high TB prevalence (e.g., Southeast Asia, Sub-Saharan Africa).
  - **Secondary Focus:** Emergency departments in urban hospitals requiring rapid screening for COVID-19 or viral pneumonia.

4. Business Processes

The system operates through a structured diagnostic workflow designed for integration into clinical environments:

1. **Image Acquisition:** High-resolution digital X-ray images are uploaded to the system via a secure portal.
2. **Automated Preprocessing:** The system automatically standardizes the input by resizing images to **224x224 pixels** and normalizing pixel values between 0 and 1, ensuring compatibility with the neural network's input requirements.

3. **Hybrid Feature Extraction:** The image is simultaneously processed through two separate neural pathways (EfficientNet and DenseNet). The resulting feature embeddings are concatenated into a single, comprehensive representation of the patient's lung health.

4. **Classification & Confidence Scoring:** The fused features pass through a dense classifier head. The system calculates a probability distribution across the four classes using a Softmax activation function.

5. **Diagnostic Reporting:** The system renders a final result, highlighting the predicted class and the model's confidence level, which is then reviewed by a medical professional.

5. Features & User Stories

Feature #1: Hybrid Fusion Classification Engine

- **Description:** An integrated classification system that merges the feature-learning capabilities of EfficientNetB3 and DenseNet121. This hybrid approach is specifically designed to handle the visual similarities between TB and Pneumonia.

- **User Story 1:** *As a radiologist, I want the system to analyze my patient's X-ray using multiple AI architectures simultaneously so that I can receive a more robust and reliable diagnostic suggestion than a single-model system would provide.*

Feature #2: Automated Diagnostic Triage

- **Description:** A real-time analysis tool that flags high-risk images (COVID-19 or TB) immediately upon upload.

- **User Story 2:** *As a triage nurse in a busy clinic, I want the system to automatically flag images that show high probability for contagious diseases like COVID-19 so that I can isolate these patients and prioritize their files for immediate doctor review.*

Feature #3: Detailed Prediction Probabilities

- **Description:** Instead of a simple "Pass/Fail" result, the system provides the exact confidence percentage for each of the four possible conditions.

- **User Story 3:** *As a general practitioner, I want to see the confidence scores for all categories so that if the model is only 51% sure between TB and Pneumonia, I know I must order a sputum test for definitive verification.*

Feature #4: Latent Space & t-SNE Analysis

- **Description:** A research-oriented feature that visualizes how the model "sees" the data, plotting image features in a 2D space to show clear separation between healthy and diseased lungs.

- **User Story 4:** *As a medical researcher, I want to see a t-SNE visualization of our patient population's X-rays so that I can identify trends in how different strains of viral pneumonia cluster compared to traditional TB cases.*

Feature #5: Performance-Optimized Training Logic

- **Description:** The training pipeline uses **AdamW optimization** and **Class Weights** to ensure that less frequent but highly dangerous diseases like TB are given equal diagnostic weight during the model's learning phase.

- **User Story 5:** *As a data scientist, I want the training process to account for class imbalance so that the final model doesn't become biased toward "Normal" results simply because there are more healthy patients in the dataset.*

6. Authorization Matrix

Role	Access Level	Permissions
Guest/Viewer	Restricted	Can only view general system statistics and public health trends.
Lab Technician	Intermediate	Can upload X-ray images, perform preprocessing, and manage patient IDs.
Specialist (Doctor)	Advanced	Can view full diagnostic results, confidence scores, and sign off on reports.
Data Scientist/Researcher	Technical	Can access t-SNE plots, performance metrics, and retrain the model.
System Admin	Full	Can manage user accounts, hardware resources, and data security protocols.

7. Assumptions

- **Image Quality:** It is assumed that input X-rays are of standard diagnostic quality and follow the Posterior-Anterior (PA) view format.
- **Infrastructure:** The system assumes access to a cloud or local environment with GPU acceleration (e.g., NVIDIA T4 or better) to maintain an inference time of less than 200ms per image.
- **Data Security:** All shared data is assumed to be anonymized in compliance with medical data privacy standards (e.g., HIPAA or GDPR) before being processed by the AI.
- **Deployment:** The model will be used as a "Human-in-the-loop" system where the final clinical decision always rests with a qualified medical professional.